CONCEPT MAP

COMPARISON OF DATA MODELS

Course code: CSA0503
Course Name: Database Management systems.

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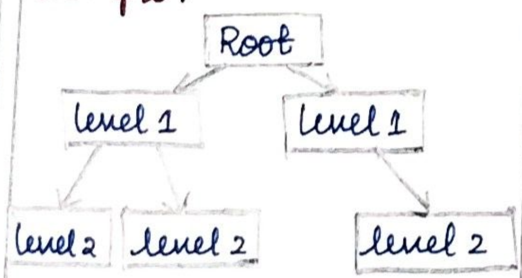
Hierarchial data model

Structure: Data is organized in a tree-like structure with one root and multiple levels.

Example:

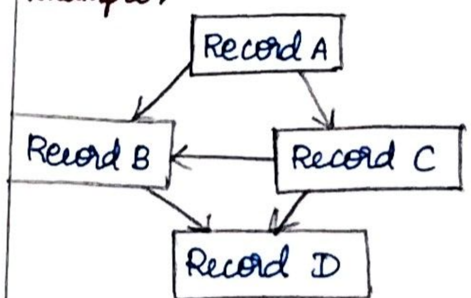


Properties:
- Parent-child relationship
- One to many (1:M) relationship.
- Example: IBM information

Network data model

Structure: Data is organized as a graph with records connected by links (owners).

Example:



Properties:
- Records are connecting using pointers (links).
- A child can have many parents
- Ex: CODASYL database

Relational Data model

Structure: Data is organized in tables (relations) consisting of rows and columns.

Example: Table: Student

Roll.no.	Name	Dept
101	Ravi	CSE
102	Anu	ECE
103	Kiran	ME

Properties:
- Data represented in tables (relations).
- Rows represent tuples (records).
- MySQL, Oracle.

Course Outcome - 1 Assessment Tool - 2 Concept Mapping.

EER Model
(Extended Entity Relationship)

An extension of the Basic ER model that adds more expressive power to model real-world scenarios more precisely.

Basic ER Model

- Uses entities, attributes and relationships to represent data.
- Supports strong entity sets and relationship sets.

Benefits of EER Model:

- Models real-world scenarios more accurately
- Supports inheritance
- Reduces redundancy and improves data integrity.

Main Extensions

Generalization

Combines multiple lower-level entity sets with similar attributes into a higher-level (super class) entity set.

Specialization

Divides a higher-level entity set (superclass) into multiple lower entity sets (subclasses) based on specific characteristics.

Aggregation

Treats a relationship set as a higher-level entity so that it can participate in another relationship.

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